**MECE E4612 — Sustainable Manufacturing**

Columbia University, Department of Mechanical Engineering

# Design for Disassembly of Disposable Electronic Cigarettes: A Universal Standard for Battery Recovery

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# 1 Introduction

## 1.1 Product Description

**[TODO: Insert product photo here — full device, front and back views.]**

The product selected for this project is a disposable electronic cigarette (e-cigarette), also known as a single-use vape. **[TODO: Specify exact brand, model, and where it was purchased. Include approximate retail price (must be under $20).]**

A disposable e-cigarette is a self-contained, battery-powered device that heats a liquid solution ("e-liquid") containing nicotine, propylene glycol, vegetable glycerin, and flavorings to produce an inhalable aerosol. The device is designed for single use: once the e-liquid is depleted or the battery is exhausted (typically after 200–800 puffs), the entire unit is discarded.

## 1.2 Functionality Analysis

The operating principle is straightforward: when the user inhales, an airflow sensor on the printed circuit board (PCB) detects the pressure change and activates the battery circuit. The lithium-ion battery then delivers current to a resistive heating coil (the atomizer), which rapidly heats to approximately 200–300 °C. The coil is wrapped around an absorbent cotton wick saturated with e-liquid. The heat vaporizes the liquid, producing an aerosol that the user inhales through the mouthpiece.

Key functional subsystems:

1. **Power supply**: Lithium-ion or lithium-polymer cell (typically 280–1000 mAh, 3.7 V nominal)
2. **Control electronics**: PCB with airflow sensor, battery protection IC, and LED indicator
3. **Atomization**: Resistive coil (kanthal or nichrome wire) wound around cotton wick
4. **Fluid storage**: Absorbent reservoir holding 2–15 mL of e-liquid
5. **Enclosure**: Outer housing (aluminum or plastic) with integrated mouthpiece

## 1.3 Scope of Analysis

This project analyzes the **entire product** rather than a subassembly. The device consists of **[TODO: specify exact count, e.g., 6–7]** distinct parts, satisfying the project requirement of at least five parts. Among these parts, we identify at least three different material types including two metallic materials (Section 2), and at least three parts requiring multiple manufacturing processes (Section 2.3).

## 1.4 Designed Usage and Durability

Disposable e-cigarettes are engineered for a single-use lifecycle. The designed usage is approximately **[TODO: specify puff count from packaging]** puffs, after which the device is intended to be discarded. There is no designed mechanism for recharging, refilling, or servicing. The battery, which Oxford and UCL researchers demonstrated is capable of over 450 charge–discharge cycles [8], is used exactly once before disposal.

### 1.5 Motivation: The E-Waste Crisis

The environmental impact of disposable e-cigarettes is substantial and growing:

- Over 840 million vaping devices are discarded globally each year, at a rate of approximately 5.7 devices per second [4].
- An estimated 30 tons of lithium from e-cigarette batteries enters landfills annually—enough to produce approximately 3,350 electric vehicle batteries [5].
- Improperly discarded lithium-ion batteries cause waste facility fires costing at least $95 million per year in the United States [7].
- A 2024 study found that 52.9% of U.S. e-cigarette users aged 15–24 dispose of devices in regular trash [6].

This project investigates how Design for Disassembly (DfD) principles can be applied to redesign the disposable e-cigarette, with the central goal of enabling efficient battery recovery and recycling through a proposed universal removable battery standard.

## 2 Materials and Manufacturing Processes

### 2.1 Part Identification and Disassembly

**[TODO: Insert photos of disassembly process from makerspace session (3/2/26).]**

The Air Bar NEX 3K was sealed via press-fit (mouthpiece into aluminum shell body). Disassembly required an X-acto knife to separate the mouthpiece from the body and was **destructive**—the press-fit joint was permanently deformed during removal. Reaching the battery required approximately **10 distinct steps** and took **4–5 minutes**, confirming that the current design is not intended for disassembly or component recovery.

The device was disassembled into the following distinct parts:

Table 1: Part inventory of the disassembled e-cigarette.

| # | Part | Material | Weight (g) |
|---|---|---|---|
| 1 | Battery cell (13450) | Li-ion (Co, Ni, Li, graphite anode) | 12.0 |
| 2 | Outer housing | Aluminum (anodized) | 13.0 |
| 3 | Atomizer coil + wick | Kanthal/nichrome wire, cotton | —[a] |
| 4 | PCB assembly | FR4, copper, solder, ICs | 2.0 |
| 5 | Mouthpiece | Black plastic (PP) | —[b] |
| 6 | E-liquid reservoir (incl. atomizer) | Cotton / plastic absorbent, e-liquid | 10.0 |
| 7 | Seals, inner chassis, wires & insulators | Silicone, white plastic, copper wire | 15.0[b] |
| | **Total** | | 52.0 |

[a] Atomizer coil + wick weight included in the reservoir combined weight (row 6).
[b] Mouthpiece, inner chassis (white plastic frame), wires, and seals were not individually weighed. The combined remainder is 15.0 g (52.0 – 12.0 – 13.0 – 2.0 – 10.0).

## 2.2 Material Identification

Materials were identified through visual inspection, magnet testing, and examination of manufacturer markings. The battery wrapper was labeled with chemistry and capacity data (13450, 3.7 V, 650 mAh, 2.405 Wh). The housing was confirmed as aluminum by its light weight (13 g for the full shell), non-magnetic behavior, and characteristic silvery appearance beneath the green anodized finish at the cut edge (wall thickness: 0.7 mm). No recycling symbols were present on any part—only branding.

### 2.2.1 Metallic Materials

The product contains at least **two distinct metallic materials**, satisfying the project requirement:

1. **Aluminum alloy** (housing): Confirmed by light weight (13 g for full shell at 0.7 mm wall thickness), non-magnetic behavior, silvery appearance at cut edge beneath green anodized surface finish.
2. **Kanthal / Nichrome** (atomizer coil): Resistance heating wire embedded in the cotton wick assembly. Coil was pressed into the juice-soaked foam reservoir.
3. **Copper** (PCB traces): Visible as reddish-orange traces on the circuit board.
4. **Solder** (PCB joints): Tin-lead (Sn-Pb) or lead-free (Sn-Ag-Cu) solder. **[TODO: Identify type.]**
5. **Battery cathode metals**: Cobalt, nickel, manganese, and lithium in the Li-ion cell cathode. Graphite anode. Aluminum and copper current collectors.

### 2.2.2 Non-Metallic Materials

1. **Plastics**: Black polypropylene (PP) mouthpiece and white plastic inner chassis frame (likely PC or ABS). No recycling codes were marked on any plastic parts.
2. **FR4 substrate**: Fiberglass-reinforced epoxy laminate (PCB base material). Contains brominated flame retardants.
3. **Cotton**: Organic cotton wick in the atomizer assembly and absorbent reservoir material.
4. **Silicone**: O-rings and sealing gaskets.
5. **Battery electrolyte**: Lithium salt in organic solvent (sealed inside the cell).

**[TODO: Insert close-up photos of each material after identification tests.]**

## 2.3 Manufacturing Process Analysis

At least **three parts require multiple manufacturing processes**, satisfying the project requirement:

#### 2.3.1 Battery Cell (3+ processes)

1. Electrode slurry coating onto current collector foils (Al for cathode, Cu for anode)
2. Calendering (compressing electrode layers)
3. Cell stacking or jellyroll winding
4. Electrolyte filling and sealing
5. Spot welding of tabs to terminals
6. Wrapping in protective sleeve (heat-shrink PVC or PET)

#### 2.3.2 Outer Housing (3+ processes)

1. Aluminum impact extrusion from slug (stadium/rounded-rectangle cross-section, 0.7 mm wall)
2. Anodizing for green surface finish
3. Pad printing for branding/graphics

#### 2.3.3 PCB Assembly (3+ processes)

1. PCB fabrication: copper lamination, photolithographic etching, drilling
2. Solder paste application (stencil printing)
3. Surface-mount technology (SMT) component placement
4. Reflow soldering
5. **[TODO: Through-hole soldering for any leaded components]**
6. Conformal coating or solder mask application

#### 2.3.4 Other Parts

- **Mouthpiece**: Injection molding (single process)
- **Atomizer coil**: Wire drawing → coil winding → assembly with wick (2–3 processes)
- **Seals**: Compression molding or injection molding of silicone

### 2.4 Use Phase, End of Life, and Transportation

**[TODO: Estimate or specify:]**

- **Use phase**: Energy consumed per device (battery capacity × voltage = 2.405 Wh, per battery marking: 650 mAh at 3.7 V). No maintenance, no consumables beyond initial e-liquid.
- **End of life**: Currently 100% landfill/incineration for most users. No take-back program from manufacturers. Battery is a hazardous waste item requiring special handling.
- **Transportation**: Manufactured in Shenzhen, China. Sea freight to US East Coast: ∼20,000 km. At 52 g per unit, a standard 20-ft container (∼22,000 kg payload) holds ∼420,000 devices.

# 3 Design for Disassembly Analysis

## 3.1 Overview of DfD Principles

Design for Disassembly (DfD) is an environmentally conscious design practice that incorporates ease of disassembly into product design at the earliest stages. The goal is to enable efficient separation of parts and materials at end of life for recycling, reconditioning, refurbishment, remanufacturing, repair, and service [3, 2].

The following DfD principles are most relevant to the redesign of disposable electronic cigarettes.

## 3.2 Principle 1: Use Reversible Fasteners

**Current design**: The housing is sealed using **[TODO: ultrasonic welding / adhesive bonding / press-fit with interference — describe what was observed during disassembly]**. Disassembly required **[TODO: cutting / prying / destructive force]**, damaging both the housing and internal components.

**DfD redesign**: Replace permanent joining with **snap-fit connectors**. A snap-fit housing consists of flexible cantilever beams with catches that deflect during assembly and lock into position, allowing non-destructive separation by applying localized force at the release points. This enables repeated assembly/disassembly without tools.

**[TODO: Insert diagram: current joining method vs. proposed snap-fit design.]**

## 3.3 Principle 2: Standardize and Minimize Fasteners

**Current design**: **[TODO: Describe fastener situation — typically no screws at all in disposable vapes; everything is press-fit or welded.]**

**DfD redesign**: If screws are needed (e.g., for battery door retention), use a single standard screw type (e.g., Phillips M2) throughout the product. Minimize total fastener count to reduce disassembly time and tool requirements.

## 3.4 Principle 3: Design Modular Architecture

**Current design**: All components are integrated into a single sealed unit. The battery, PCB, atomizer, and reservoir are assembled sequentially and cannot be individually accessed or removed.

**DfD redesign**: Separate the product into three self-contained modules:

1. **Battery module**: Removable, standardized lithium-ion cell in a snap-fit compartment
2. **Atomizer/reservoir module**: Replaceable cartridge containing the coil, wick, and e-liquid
3. **Body/electronics module**: Housing with integrated PCB, mouthpiece, and electrical contacts

This modular architecture allows each module to be directed to the appropriate recycling stream at end of life, and enables battery reuse across multiple device lifecycles.

## 3.5 Principle 4: Make Hazardous Components Most Accessible

**Current design**: The lithium-ion battery is buried in the center of the device, surrounded by the reservoir and housing. Accessing it requires complete destructive disassembly.

**DfD redesign**: Position the battery compartment at the **base of the device** with a snap-fit or twist-lock door. The battery should be the **first component removable** without any tools, following a linear disassembly path. This is critical because:

- Lithium-ion batteries are classified as hazardous waste
- Batteries in the waste stream cause fires at a cost of $95M/year [7]
- Easy removal encourages consumer participation in battery recycling

### 3.6 Principle 5: Mark Materials for Identification

**Current design**: **[TODO: Describe whether any material markings exist. Typically, disposable vapes have no recycling symbols or material identification codes on any component.]**

**DfD redesign**: Apply standard material identification markings:

- ISO 11469 plastic identification codes on all plastic parts (e.g., >PP<, >PC<)
- Metal type stamps on metallic housing components
- Battery chemistry labeling (Li-ion, capacity, voltage)
- QR code linking to recycling instructions (aligned with EU Batteries Regulation requirements [1])

### 3.7 Principle 6: Use Compatible Materials

**Current design**: Mixed material construction — aluminum housing bonded to plastic components, copper PCB attached with solder, different plastic types used without separation features.

**DfD redesign**: Group compatible materials within modules to avoid cross-contamination during recycling:

- Mono-material plastic housing (single polymer type, e.g., PP)
- All electronics (PCB, wiring) contained in one separable module
- Battery isolated with its own current collectors and contacts
- Avoid coatings or surface treatments that prevent material recyclability

### 3.8 Principle 7: Minimize Disassembly Operations

**Current design**: **[TODO: Count the number of operations required to fully disassemble the current product. Time the process. Note tools required.]**

**DfD redesign target**:

- Battery removal: **1 step**, 0 tools, <5 seconds
- Full disassembly to individual modules: **3 steps**, 0 tools, <30 seconds
- Complete disassembly to individual parts: ≤**5 steps**, 0–1 standard tools

### 3.9 Disassembly Comparison

Table 2: Disassembly comparison: current design vs. DfD redesign.

| Metric | Current Design | DfD Redesign |
|---|---|---|
| Steps to remove battery | **[TODO: X]** | 1 |
| Total disassembly steps | **[TODO: X]** | ≤5 |
| Tools required | **[TODO: pliers, knife]** | None (hand only) |
| Disassembly time | **[TODO: X min]** | <30 sec |
| Destructive? | Yes | No |
| Battery reusable after removal | No | Yes |
| Material recovery rate | ∼0% | >90% (target) |

**[TODO: Insert side-by-side photos or diagrams showing the current vs. redesigned disassembly sequence.]**

## 4 Life Cycle Assessment

### 4.1 Methodology

Life Cycle Assessment (LCA) was conducted following **[TODO: ISO 14040/14044 framework]** to compare the environmental impacts of the reference product (current disposable e-cigarette design) against the DfD-redesigned concept.

**Software**: **[TODO: Specify: SimaPro, OpenLCA, GaBi, or other LCA tool used.]**

**Database**: **[TODO: Specify: Ecoinvent, USLCI, or other LCI database.]**

**Functional unit**: One disposable e-cigarette device delivering ∼3,000 puffs of nicotine aerosol (Air Bar NEX 3K rated capacity).

**System boundary**: Cradle-to-grave, including:

- Raw material extraction and processing
- Manufacturing and assembly
- Transportation (manufacturing site to point of sale)
- Use phase (negligible environmental impact beyond embodied energy)
- End-of-life (landfill for reference; recycling pathway for DfD concept)

## 4.2 Simplified Bill of Materials (SBOM)

Table 3: Simplified Bill of Materials for LCA input.

| Component | Material (LCI Entry) | Mass (g) |
|---|---|---|
| Battery cell (13450) | Lithium-ion battery, rechargeable (3.7 V, 650 mAh) | 12.0 |
| Housing | Aluminium, wrought alloy (anodized) | 13.0 |
| PCB assembly | Printed circuit board, surface mounted | 2.0 |
| Atomizer coil + wick | Kanthal/nichrome wire, cotton | incl. below |
| Mouthpiece | Polypropylene, injection molded | —[a] |
| E-liquid reservoir (incl. atomizer) | Cotton fibre, plastic absorbent, e-liquid | 10.0 |
| Inner chassis, seals, wires | PC/ABS, silicone rubber, copper wire | —[a] |
| **Total** | | 52.0 |

[a] Mouthpiece, inner chassis, seals, and wires were not individually weighed. Combined remainder: $52.0 - 12.0 - 13.0 - 2.0 - 10.0 = 15.0\,\text{g}$.

## 4.3 Reference Product LCA (Current Design)

**[TODO: Insert LCA results for the current disposable e-cigarette design. Include:]**

### 4.3.1 Impact Categories

**[TODO: Present results for at least these impact categories:]**

- Global Warming Potential (GWP, kg $CO_2$-eq)
- Acidification Potential (kg $SO_2$-eq)
- Eutrophication Potential (kg $PO_4$-eq)
- Abiotic Resource Depletion (kg Sb-eq)
- Human Toxicity Potential (kg 1,4-DCB-eq)
- **[TODO: Additional categories as appropriate]**

### 4.3.2 Life Cycle Stage Breakdown

**[TODO: Present results broken down by life cycle stage:]**

- Materials extraction and processing
- Manufacturing
- Transportation
- Use phase
- End of life (landfill scenario)

### 4.4 DfD Concept LCA

**[TODO: Insert LCA results for the DfD-redesigned e-cigarette. Key differences in modeling:]**

- **End-of-life**: Battery directed to lithium-ion recycling (hydrometallurgical recovery). Housing directed to plastic or metal recycling. PCB directed to e-waste processing.
- **Material credits**: Apply recycling credits for recovered lithium, cobalt, nickel, copper, and aluminum.
- **Additional manufacturing**: Account for snap-fit features, battery door, material marking — marginal increase in manufacturing impact.
- **Battery reuse scenario**: If battery is reused across multiple device lifecycles, allocate battery impacts across number of uses.

### 4.5 Identification of Greatest Impacts

**[TODO: Based on the LCA results, identify:]**

1. The **SBOM input** (material) with the greatest environmental impact
2. The **impact category** most significantly affected
3. The **life cycle stage** contributing most to overall impacts

**[TODO: These findings drive the alternative concept design in Section 5.]**

## 5 Alternative Concept: The Universal Removable Battery Standard

Based on the LCA results identifying **[TODO: specify greatest impact — likely battery materials and end-of-life landfilling]** as the dominant environmental impact, this section presents an alternative concept designed to minimize these impacts: a **universal removable battery standard** for electronic cigarettes.

### 5.1 Design Rationale

The Oxford/UCL study demonstrated that lithium-ion cells in disposable e-cigarettes can sustain over 450 charge–discharge cycles [8], yet are discarded after a single use. The core insight driving this concept is that the battery—the most environmentally impactful component—can be **decoupled from the consumable portion** of the device and **standardized across manufacturers**.

This approach draws on precedent from:

- The AA/AAA battery standard (IEC 60086), which enables interchangeability across thousands of products from hundreds of manufacturers
- The EU Batteries Regulation (Article 11), which mandates removable, replaceable batteries in portable electronics by February 2027 [1]
- The Fairphone modular smartphone design, which demonstrates commercial viability of DfD consumer electronics

## 5.2 Proposed Standard Specification

The standard defines **three battery form factors** covering the full range of disposable e-cigarette capacities:

Table 4: Proposed universal e-cigarette battery standard — three sizes.

| Size Class | Diameter (mm) | Length (mm) | Capacity (mAh) | Use Case |
|---|---|---|---|---|
| VC-S (Small) | **[TODO: 8–10]** | **[TODO: 30–35]** | 280–400 | Compact / low-puff devices |
| VC-M (Medium) | **[TODO: 10–13]** | **[TODO: 35–45]** | 400–650 | Standard disposables |
| VC-L (Large) | **[TODO: 13–15]** | **[TODO: 45–55]** | 650–1000 | High-capacity devices |

**[TODO: Insert dimensional drawing of each battery size class with tolerances.]**

### 5.2.1 Mechanical Interface

- **Battery compartment**: Located at the base of the device, accessible via a twist-lock or snap-fit door
- **Insertion method**: Drop-in cylindrical cell with keyed orientation (polarity alignment via asymmetric tab)
- **Retention**: Spring-loaded contact at one end provides both electrical connection and mechanical retention
- **Removal**: Single-handed operation — press release tab and slide battery out

### 5.2.2 Electrical Interface

- **Contacts**: Gold-plated spring contacts at positive and negative terminals
- **Nominal voltage**: 3.7 V (standard Li-ion)
- **Protection**: Each battery cell includes an integrated protection circuit (overcharge, overdischarge, short-circuit)
- **Charging**: Standardized USB-C charging port on the device body (not on the battery)

### 5.2.3 Identification and Labeling

- **Color coding**: Each size class has a distinct color band (e.g., Blue = S, Green = M, Orange = L)
- **QR code**: Links to recycling instructions, battery chemistry, manufacturer, and date of production (per EU Batteries Regulation requirements)
- **Material marking**: Battery sleeve marked with Li-ion recycling symbol and chemistry code

### 5.3 Device Body Redesign

**[TODO: Insert exploded view diagram or CAD render of the redesigned device.]**

Key design changes to the device body:

1. **Housing material**: Switch from aluminum to **mono-material recyclable plastic** (polypropylene or recycled PET) to eliminate mixed-material contamination and reduce embodied energy
2. **Snap-fit assembly**: Two-piece clamshell housing with integrated snap-fit clips — no adhesives, no ultrasonic welding, no screws
3. **Battery compartment**: Molded receptacle at the base with spring contacts and twist-lock door
4. **Modular atomizer cartridge**: Slide-in cartridge containing coil, wick, and e-liquid reservoir — separate recycling stream from battery and housing
5. **Standardized PCB mounting**: PCB secured with press-fit standoffs, removable without tools

### 5.4 Impact Reduction Mechanisms

1. **Battery lifecycle extension**: Each battery reused across 10–450+ device lifecycles instead of 1, reducing per-use lithium demand by 90–99.8%
2. **Elimination of waste stream fires**: Batteries removed before disposal, preventing lithium-ion fires in waste facilities
3. **Material recovery**: >90% battery material recovery through established Li-ion recycling pathways (vs. ∼0% currently)
4. **Reduced raw material extraction**: Less cobalt, nickel, and lithium mining per functional unit
5. **Simplified recycling**: Mono-material housing enables single-stream plastic recycling; modular separation eliminates manual sorting

### 5.5 LCA of Alternative Concept

**[TODO: Run LCA for the alternative concept and present results. Key modeling differences from DfD concept:]**

- Battery impact allocated across **[TODO: X]** use cycles (not just 1)
- Housing material changed from aluminum to PP (lower embodied energy)
- End-of-life credits for battery recycling and plastic recycling
- Additional: USB-C charging port and spring contacts (marginal)

# 6 Results and Discussion

## 6.1 LCA Comparison

**[TODO: Present the three-way LCA comparison (Reference vs. DfD Concept vs. Alternative Concept) across all impact categories. Use bar charts or tables.]**

Table 5: LCA results comparison across three design concepts.

| Impact Category | Reference | DfD Concept | Alternative |
|---|---|---|---|
| GWP (kg $CO_2$-eq) | **[TODO: X]** | **[TODO: X]** | **[TODO: X]** |
| Acidification (kg $SO_2$-eq) | **[TODO: X]** | **[TODO: X]** | **[TODO: X]** |
| Eutrophication (kg $PO_4$-eq) | **[TODO: X]** | **[TODO: X]** | **[TODO: X]** |
| Resource Depletion (kg Sb-eq) | **[TODO: X]** | **[TODO: X]** | **[TODO: X]** |
| Human Toxicity (kg 1,4-DCB-eq) | **[TODO: X]** | **[TODO: X]** | **[TODO: X]** |

## 6.2 Disassembly Performance

Table 6: Disassembly performance metrics.

| Metric | Reference | DfD Concept | Alternative |
|---|---|---|---|
| Battery removal steps | ∼10 | 1 | 1 |
| Battery removal time | 4–5 min | <5 sec | <5 sec |
| Full disassembly steps | ∼10 | ≤5 | ≤5 |
| Full disassembly time | 4–5 min | <30 sec | <30 sec |
| Tools required | X-acto knife | None | None |
| Destructive? | Yes | No | No |

## 6.3 Material Recovery Analysis

Table 7: Estimated material recovery rates by design concept.

| Material | Reference | DfD Concept | Alternative |
|---|---|---|---|
| Lithium | ∼0% | >90% | >95% (reuse + recycle) |
| Cobalt/Nickel | ∼0% | >90% | >95% |
| Copper | ∼0% | ∼80% | ∼85% |
| Aluminum | ∼0% | ∼85% | N/A (PP housing) |
| Plastics | ∼0% | ∼50% | >80% (mono-material) |

## 6.4 Cost Implications

**[TODO: Analyze the cost implications of the redesign:]**

- **Additional manufacturing cost**: Snap-fit tooling, battery door mechanism, spring contacts, material marking. Estimate: **[TODO: $X per unit increase.]**

- **Material savings**: Battery reuse reduces per-device battery cost. If battery is reused 10 times, battery cost per device drops by ∼90%.
- **Regulatory compliance**: DfD design preemptively meets EU Batteries Regulation 2027 requirements, avoiding future redesign costs.
- **Waste management savings**: Reduced fire risk and simplified recycling lower municipal waste processing costs.

### 6.5 Discussion

**[TODO: Interpret the results. Consider:]**

- Which impact categories show the greatest reduction from the alternative concept?
- How does the battery reuse assumption (number of cycles) affect the results?
- What are the barriers to adoption? (manufacturer resistance, consumer behavior, cost, standardization governance)
- How does this compare to the outright ban approach (UK, Australia)?
- What role does the EU Batteries Regulation play as a forcing function?
- Limitations: assumptions in LCA modeling, idealized recycling rates, consumer participation uncertainty

## 7 Conclusions

**[TODO: Write conclusions based on final results. The following structure is suggested:]**

This project analyzed the environmental impact of disposable electronic cigarettes and applied Design for Disassembly principles to redesign the product for improved end-of-life management. The key findings are:

1. **[TODO: Finding 1: Current disposable e-cigarettes are designed as sealed, single-use units that prevent any material recovery. The LCA showed that [greatest impact category] is dominated by [SBOM input / life cycle stage].]**
2. **[TODO: Finding 2: Applying DfD principles—particularly snap-fit assembly, modular architecture, and prioritized battery accessibility—reduced disassembly time from X minutes to under 30 seconds and eliminated the need for destructive disassembly.]**
3. **[TODO: Finding 3: The proposed Universal Removable Battery Standard, defining three standardized battery sizes with mechanical and electrical interface specifications, could reduce per-device environmental impacts by X% across [impact categories] by enabling battery reuse across multiple device lifecycles.]**
4. **[TODO: Finding 4: The alternative concept with mono-material PP housing and modular design achieves material recovery rates exceeding 90% for critical metals, compared to approximately 0% for the current design.]**

The proposed standard aligns with the EU Batteries Regulation (2023/1542), which will require removable and replaceable batteries in all portable electronics by February 2027. Early adoption of such a standard would provide manufacturers with a compliance pathway while addressing the urgent environmental and safety concerns associated with the current generation of disposable e-cigarettes.

### 7.1 Limitations

**[TODO: Discuss limitations:]**

- LCA assumptions and data quality
- Idealized recycling rate assumptions
- Consumer behavior uncertainty (will users actually remove batteries?)
- Standardization governance (who administers the standard?)
- Potential for the standard to extend the lifecycle of nicotine products (public health consideration)

### 7.2 Future Work

**[TODO: Suggest future directions:]**

- Prototype fabrication and physical disassembly testing
- Consumer behavior studies on battery removal compliance
- Detailed cost–benefit analysis for manufacturers
- Engagement with standards bodies (IEC, ISO) for formal standardization
- Extension to rechargeable/refillable e-cigarette designs

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